

KITE FAMILIES WITH $\Theta(n^3)$ FAILURE OF THE ZHOU–WANG–CHAI DISTANCE-LAPLACIAN BOUND

ABSTRACT. Let $\partial_1^L(G) \geq \cdots \geq \partial_n^L(G) = 0$ be the distance Laplacian eigenvalues of a connected graph G , and write $U_r(G) = \sum_{i=1}^r \partial_i^L(G)$. Zhou, Wang, and Chai proposed the bound

$$U_r(G) \leq W(G) + \binom{r+2}{3}.$$

Huang [3] has already shown that this bound fails for $K_{1,3}$ at $r = 2$ and for $K_{1,4}$ at $r = 3$, so the unrestricted conjecture is false; our point is that the failure is neither sporadic nor small. We show that failure persists throughout an infinite unicyclic family. If H_n is the kite graph obtained from the path $v_1 v_2 \cdots v_n$ by adding $v_1 v_3$, then every H_n , $n \geq 5$, violates the bound at $r = n - 2$, with gap at least $(3n - 14)/2$. Along $n = 3m$, a second choice $r = 2m$ gives a violation of order n^3 . The graph H_5 has minimum possible order among unicyclic counterexamples.

1. THE COUNTEREXAMPLES

For a connected graph G , let $D(G) = (d_G(u, v))$ be its distance matrix, let

$$\tau_G(v) = \sum_{u \in V(G)} d_G(u, v), \quad W(G) = \frac{1}{2} \sum_{v \in V(G)} \tau_G(v),$$

and let

$$D^L(G) = \text{diag}(\tau_G(v) : v \in V(G)) - D(G)$$

be its distance Laplacian matrix [1]. Write

$$\partial_1^L(G) \geq \cdots \geq \partial_n^L(G) = 0, \quad U_r(G) = \sum_{i=1}^r \partial_i^L(G),$$

and define the excess

$$\Delta_r(G) = U_r(G) - W(G) - \binom{r+2}{3}.$$

Zhou, Wang, and Chai [5] conjectured that $\Delta_r(G) \leq 0$ for every connected graph and every $2 \leq r \leq n$. A recent preprint of Huang [3] classifies the diameter-two case and finds precisely the exceptional pairs $(K_{1,3}, 2)$ and $(K_{1,4}, 3)$. Our contribution is an infinite family inside the more restrictive class of unicyclic graphs.

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For $n \geq 4$, let H_n have vertex set $\{v_1, \dots, v_n\}$ and edge set

$$E(H_n) = \{v_i v_{i+1} : 1 \leq i < n\} \cup \{v_1 v_3\}.$$

Thus H_n is the standard kite graph $\text{Ki}_{n,3}$: a triangle with a pendant path [4].

Theorem 1. *For every $n \geq 5$,*

$$\Delta_{n-2}(H_n) \geq \frac{3n-14}{2} > 0.$$

Moreover, if $n = 3m$ with $m \geq 4$, then

$$\Delta_{2m}(H_{3m}) \geq \frac{5m^3 - 12m^2 - 17m + 12}{6} > 0.$$

In particular, the violation is $\Theta(n^3)$ along the subfamily H_{3m} .

Proof. For every $x \in \mathbb{R}^{V(G)}$,

$$(1) \quad x^\top D^L(G)x = \sum_{\{u,v\} \subseteq V(G)} d_G(u,v)(x_u - x_v)^2.$$

Hence $D^L(G)$ is positive semidefinite and, since G is connected, $\ker D^L(G) = \text{span}\{\mathbf{1}\}$. Therefore

$$(2) \quad \partial_{n-1}^L(G) = \min_{\substack{x \neq 0 \\ x \perp \mathbf{1}}} \frac{x^\top D^L(G)x}{x^\top x}.$$

Only the distances $d(v_1, v_j)$, $3 \leq j \leq n$, decrease when $v_1 v_3$ is added to P_n , and each decreases by one. Thus

$$(3) \quad W(H_n) = W(P_n) - (n-2) = \binom{n+1}{3} - (n-2).$$

Take $x = e_3 - e_4$. The relevant transmissions are

$$\tau_{H_n}(v_3) = 2 + \frac{(n-3)(n-2)}{2} = \frac{n^2 - 5n + 10}{2}$$

and

$$\tau_{H_n}(v_4) = 5 + \frac{(n-4)(n-3)}{2} = \frac{n^2 - 7n + 22}{2}.$$

Since $x \perp \mathbf{1}$ and $d(v_3, v_4) = 1$, equations (1)–(2) give

$$\partial_{n-1}^L(H_n) \leq \frac{\tau_{H_n}(v_3) + \tau_{H_n}(v_4) + 2}{2} = \frac{n^2 - 6n + 18}{2}.$$

Also $\text{tr } D^L(H_n) = 2W(H_n)$, so

$$U_{n-2}(H_n) = 2W(H_n) - \partial_{n-1}^L(H_n).$$

Using (3),

$$\begin{aligned}\Delta_{n-2}(H_n) &\geq W(H_n) - \binom{n}{3} - \frac{n^2 - 6n + 18}{2} \\ &= \frac{3n - 14}{2}.\end{aligned}$$

This proves the first assertion.

Now let $n = 3m$ and

$$S = \{v_1, \dots, v_m\} \cup \{v_{2m+1}, \dots, v_{3m}\}.$$

Ky Fan's variational principle [2] gives

$$(4) \quad U_{2m}(H_{3m}) = \max_{Q^T Q = I_{2m}} \operatorname{tr}(Q^T D^L(H_{3m}) Q) \geq \sum_{v \in S} \tau_{H_{3m}}(v),$$

where Q ranges over real $3m \times 2m$ matrices. For the path P_{3m} ,

$$\begin{aligned}\sum_{v \in S} \tau_{P_{3m}}(v) &= 2 \sum_{i=1}^m \left[\binom{i}{2} + \binom{3m-i+1}{2} \right] \\ &= \frac{20m^3 - 2m}{3}.\end{aligned}$$

Passing from P_{3m} to H_{3m} shortens the $3m - 2$ pairs $\{v_1, v_j\}$, $3 \leq j \leq 3m$. Every such pair contributes once to the selected transmission sum through $v_1 \in S$, and $2m - 2$ of them contribute a second time because $v_j \in S$. Hence

$$(5) \quad \sum_{v \in S} \tau_{H_{3m}}(v) = \frac{20m^3 - 2m}{3} - (5m - 4) = \frac{20m^3 - 17m + 12}{3}.$$

Combining (3), (4), and (5) yields

$$\Delta_{2m}(H_{3m}) \geq \frac{5m^3 - 12m^2 - 17m + 12}{6}.$$

For $p(m) = 5m^3 - 12m^2 - 17m + 12$, one has $p(4) = 72$ and

$$p(m+1) - p(m) = 3(m+1)(5m-8) > 0 \quad (m \geq 4),$$

so the lower bound is positive. It is $\frac{5}{162}n^3 + O(n^2)$, whereas $\Delta_{2m}(H_{3m}) \leq U_{2m}(H_{3m}) \leq 2W(H_{3m}) = O(n^3)$. Thus the violation is $\Theta(n^3)$. $\square$

Remark 2. The unicyclic restriction is what separates Theorem 1 from the unrestricted statement; it is not what makes the failure infinite, nor what makes it large. Under the unrestricted formulation, the same argument with $x = e_2 - e_3$ gives

$$\Delta_{n-2}(P_n) \geq \frac{3n - 10}{2} > 0 \quad (n \geq 4).$$

Moreover, the sum $\sum_{v \in S} \tau_{P_{3m}}(v)$ evaluated in the proof of Theorem 1 already exceeds $W(P_{3m}) + \binom{2m+2}{3}$ for every $m \geq 3$, so by Ky Fan's principle [2] the paths P_{3m} by themselves violate the bound by $\Theta(n^3)$. Thus paths already

form an infinite family with cubic excess; Theorem 1 shows that failure of both orders persists after imposing unicyclicity.

Corollary 3. *The graph H_5 has minimum possible order among unicyclic counterexamples. Its distance Laplacian matrix is*

$$D^L(H_5) = \begin{pmatrix} 7 & -1 & -1 & -2 & -3 \\ -1 & 7 & -1 & -2 & -3 \\ -1 & -1 & 5 & -1 & -2 \\ -2 & -2 & -1 & 6 & -1 \\ -3 & -3 & -2 & -1 & 9 \end{pmatrix},$$

and

$$\det(xI - D^L(H_5)) = x(x - 6)(x - 8)(x^2 - 20x + 95).$$

Consequently,

$$\text{Spec}(D^L(H_5)) = \{10 + \sqrt{5}, 8, 10 - \sqrt{5}, 6, 0\},$$

so $W(H_5) = 17$ and

$$U_3(H_5) = 28 > 27 = 17 + \binom{5}{3}.$$

Proof. The displayed matrix and characteristic polynomial follow directly from the distances in H_5 . The only connected unicyclic graphs on fewer than five vertices are C_3 , C_4 , and H_4 , whose distance Laplacian spectra are, respectively,

$$\{3, 3, 0\}, \quad \{6, 6, 4, 0\}, \quad \{7, 5, 4, 0\}.$$

These spectra verify the conjectured inequality for every relevant r , while the calculation above shows that H_5 violates it at $r = 3$. $\square$

REFERENCES

- [1] M. Aouchiche and P. Hansen, *Two Laplacians for the distance matrix of a graph*, Linear Algebra Appl. **439** (2013), no. 1, 21–33. doi:10.1016/j.laa.2013.02.030.
- [2] K. Fan, *On a theorem of Weyl concerning eigenvalues of linear transformations I*, Proc. Natl. Acad. Sci. USA **35** (1949), no. 11, 652–655. doi:10.1073/pnas.35.11.652.
- [3] S. Huang, *On a distance Laplacian analog of Brouwer’s conjecture for several classes of graphs*, arXiv:2606.06945 [math.CO], 2026.
- [4] H. Lin, Z. Du, and B. Zhou, *On the first two largest distance Laplacian eigenvalues of unicyclic graphs*, Linear Algebra Appl. **546** (2018), 289–307. doi:10.1016/j.laa.2018.02.013.
- [5] Y. Zhou, L. Wang, and Y. Chai, *Brouwer type conjecture for the eigenvalues of distance Laplacian matrix of a graph*, Comput. Appl. Math. **44** (2025), Paper No. 138. doi:10.1007/s40314-025-03095-0.